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One Run, One Plug: A New Modular P&A Strategy for Faster, Safer, and Cost-Effective Abandonment

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Abstract

Objectives / Scope: This paper presents an innovative methodology for Plug and Abandonment (P&A) operations that significantly enhances efficiency, safety, and cost-effectiveness through the use of a modular single-trip cement plug placement system. The approach introduces a novel combination of a quick-disconnect tool, inflatable packer, dedicated circulation sub and sacrificial string, all deployed as a single bottom-hole assembly, enabling placement of long, tailored cement plugs in a single operation. This contrasts with conventional P&A practices, which often rely on multiple stacked cement plugs and require several dedicated trips (for placement and for tagging) leading to increased operational complexity, higher costs, and extended rig time.

Methods, Procedures, Process: The new methodology addresses these limitations by introducing modularity and design flexibility into cement plug placement customized for each well scenario. The inflatable packer provides a secure and firm mechanical base at a pre-determined depth, while the circulation sub and disconnect tool allow precise plug placement, simplified release from the work string, usage of old and rejected pipes as sacrificial string instead of sent for disposal. This configuration minimizes risks associated with traditional methods, such as plug integrity, Equivalent Circulating Density (ECD) constraints and potential formation losses, particularly in wells with weak or fractured zones.

Results, Observations, Conclusions: Field trials of this approach were conducted across multiple offshore wells in both West Africa and the Americas. In West Africa, the quick-disconnect tool was tested as a standalone system during the permanent abandonment of a 12-1/4-in. open hole. Here, 400-meter-long cement plugs were successfully placed in a single operation, replacing the conventional approach of placing two 200-meter plugs in sequence. This not only fulfilled regulatory isolation requirements but also demonstrated the tool's reliability and allowed operators to gain familiarity with its use. The trial resulted in operational simplification and an estimated cost savings of USD 1.5 million.

In the Americas, the full system—including the inflatable packer, circulation sub, and disconnect tool—was deployed in a single run for the abandonment of an 8-1/2-in. open hole. The plug base was set approximately 600 meters above the well's total depth. The entire system was assembled on a 3-1/2-in. sacrificial tubing string and deployed in one trip. The result was a successful cementing operation with no requirement to pull out of the cement, further reducing rig time and improving service quality. This operation was also completed ahead of schedule, again resulting in an estimated cost saving of USD 1.5 million.

The success of these operations demonstrates the robustness and adaptability of the system. The modularity allows for flexible plug designs tailored to specific well conditions and regulatory demands, including control over hydrostatic pressure and gas migration. The system eliminates the need for staged operations and multiple trips, directly contributing to reduced operational risk, rig time, and cost.

Novel / Additive Information: In conclusion, this innovative approach to P&A operations represents a major advancement in well abandonment practices. By combining modular design with efficient deployment, the methodology enables safer, faster, and more cost-effective operations. Its field-proven success indicates significant potential for broader application across a variety of well types and abandonment scenarios globally.

Introduction

Permanent plug and abandonment of oil and gas wells represents a critical phase in the well lifecycle, requiring strict adherence to technical and regulatory standards to ensure long-term well integrity and environmental protection. According to the operator's well abandonment procedures, specific sealing barrier requirements must be satisfied for a well to be considered permanently abandoned. For each distinct permeable zone, two independent well barriers—designated as the primary and secondary barriers—are required to prevent fluid migration to the surface or seabed. Additionally, barriers must be installed between adjacent permeable zones to prevent crossflow and must provide continuous isolation from cement to formation.

A key component of the permanent well barrier envelope (WBE) is the cement plug, which must meet the defined acceptance criteria in terms of placement depth, mechanical integrity, and length. The cement plug must be designed to deliver at least 100 meters long cement column above the uppermost potential inflow zone, where its integrity shall be verified by methods such as mechanical tagging, positive and negative (inflow) pressure testing, whenever is applicable.

Although cement plug placement is a standard procedure, it involves several technical and operational risks that may compromise the quality of the placement and, consequently, the establishment of a reliable well barrier envelope. One of the primary risks affecting plug integrity is cement slurry contamination and fluid swapping, caused by density differentials. Variations in fluid densities can cause instabilities during the displacement process, resulting in the downward migration of cement through lighter wellbore fluids and leading to contamination. These instabilities are further influenced by factors such as wellbore inclination, annular clearance between the workstring and wellbore, and the rheological properties of the fluids involved.

Another critical aspect of the operation is the removal of the workstring after plug placement. The string must be withdrawn in a controlled manner that prevents plug disturbance and ensures that the slurry within the string drains fully during retrieval, particularly in long cement columns. Laboratory testing has demonstrated that if the cement slurry has developed gel strength or exhibits high viscosity at the time of pulling, the plug is likely to become contaminated during string removal. This contamination while pulling workstring out of the cement can also be increased in well configuration where the differential pressure between fluids is limited, directly impacting on fluids hierarchy and deviated trajectories.

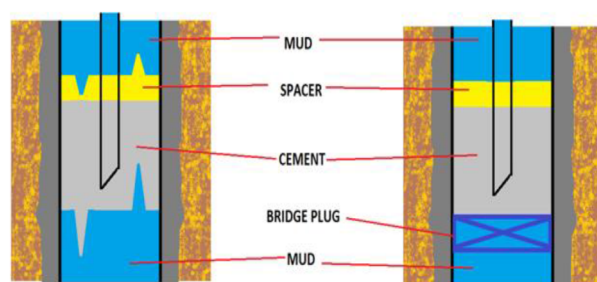


Figure 1—Cement plug contamination due to density differential

To address these challenges and optimize P&A operations, operators in recent years have conducted a comprehensive market enquiry with the goal of identifying a system that could reduce rig time, improve safety, and ensure high success rates for cement plug placement. During this market survey, multiple technologies from various service providers were evaluated. After a detailed analysis of the advantages and disadvantages of each solution, a dart-operated disconnection tool—eventually used in combination with an inflatable packer—was selected.

This system offers significant benefits in P&A operations, including the prevention of slurry contamination by allowing the tool to disconnect above the cement, leaving the slurry undisturbed. The use of an open-hole packer also provides mechanical support and prevents fluid swapping, thereby maintaining the stability of the cement column during placement and setting. In parallel, operators have focused on optimizing cement job design and execution by strictly adhering to best practices during both the planning and operational phases. These improvements include the development of optimized slurry formulations, proper displacement sequences, and robust verification procedures.

What is and how does a string quick-disconnect tool works

The quick-disconnect tool is designed with the objective of simplifying cement plug setting during well construction and abandonment operations, featuring a single-collet release mechanism that is activated by a dedicated displacement dart. The drill pipe wiper dart also offers separation and effective displacement of the cementing fluids, both critical to the success of cement plug integrity and strength development as it minimizes fluids contamination as they travel down the string.

This type of tool quickly releases a downstream tail/sacrificial pipe (usually tubing) that will be left in hole from the landing string (usually drill pipe) that will be retrieved to surface, providing a clear surface pressure indication, and eliminating the need to pull the string out of the plug length. As a result, there is no cement plug disturbance, and the risk of cement contamination is minimized. These characteristics allow for multiple zone isolation in a single cementing operation with reduced risk of string getting stuck, and effectively placement of reduced-volume kickoff or sidetrack cement plugs.

Overall, string quick-disconnect systems avoid cement slurry contamination, allow operation of the tool and disconnection from above the cement plug, meaning the slurry remains undisturbed. These help to achieve superior plug integrity and ultimately contribute to the success of the well abandonment process.

As such tool can be deployed in open or cased hole scenarios, vertical and deviated wells, and poses no limitation to cement plug length, benefits can be achieved across all types of P&A, as it also offers the benefit of allowing aggressive cement slurry design for quick compressive and gel strength development because it eliminates the need to POOH, which in result, reduces Wait On Cement times and improves cement plug tagging reliability.

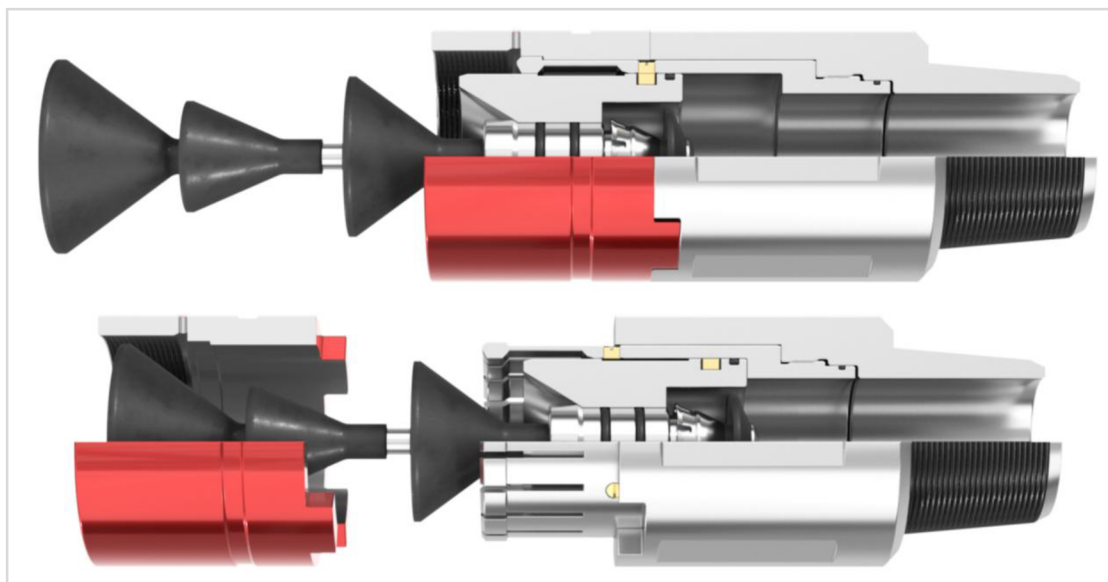


Figure 2—Example of a string quick-disconnect tool sequence of operation: Dart landed, and Collet system released

To further prevent fluid migration and mud contamination, the disconnect tool can be modularly customized to incorporate an inflatable packer into the same string, which creates a solid foundation that helps prevent the fluid-swap effect and the dart used to disconnect the tool from the workstring acts as a mechanical barrier avoiding contamination from the displacement fluid. Lastly, the bottom hole assembly can be designed as torque capable, to improve cement placement in the annular space, ensure isolation and limit potential cement returns on surface.

Since this string disconnection mechanism eliminates the need of pulling the string out of the balanced cement plug and the packer provides a supporting base below the plug, the result is basically deleting the risks of slurry contamination, eliminating as well the need for mechanical tagging: the rationale behind this aspect, implemented in the new P&A procedures, is that a reasonable certainty that the plug is going to set and achieve the mechanical/hydraulic sealing is achieved if a supporting base and sacrificial string are applied.

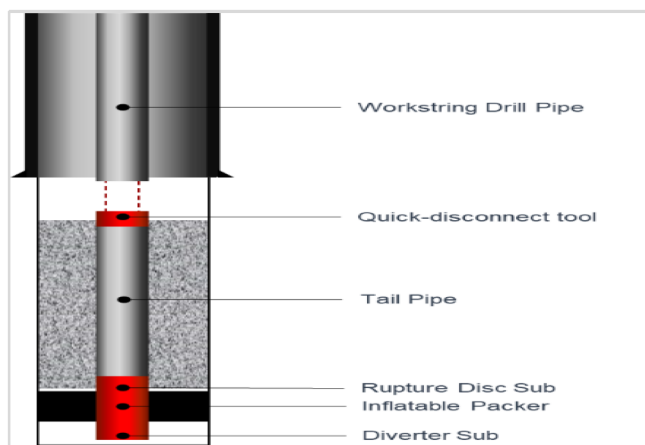


Figure 3—Schematic representation of an off-bottom P&A with a Quick-disconnect tool and Inflatable packer on single string

It is also important to highlight that the referenced system has a strong track record of field performance, with multiple successful implementations in different well environments. It offers enhanced safety by

reducing the risk of tool failures and operational hazards, and its precision in cement plug placement ensures the highest levels of well integrity.

One of the primary drivers on the selection of this approach was its ability to streamline operations by using a single-trip deployment. Traditional methods involve multiple runs in and out of the well, which increases both time and cost. With the proposed solution, the cement plug is set and released in one trip, avoiding any WOC and therefore significantly reducing rig time.

Case Study #1 – Americas

For the abandonment of the exploratory well, requiring isolation of a 8-1/2-in. open hole, operator identified the need to plug-off the wellbore, up-to the previously cemented 9 7/8-in. liner, conventionally using the bottom of the well as a proven mechanical foundation and extending the cement coverage to roughly 1,000 m. Usual P&A methodology would be to place successive balanced cement plugs, resulting in a minimum of three dedicated cementing operations to spot the plugs and execute required pre- and post-job well circulation – highly inefficient process overall. Existing alternatives on the market, (bridge plugs, mechanical packers) require dedicated runs for setting and tagging, which eventually do not provide substantial efficiency savings.

The custom and modular P&A solution that incorporates an inflatable packer into the same string as the bottom-hole disconnect assembly was jointly developed and deployed to allow for improved operational efficiencies throughout the planned P&A. The full string composition included application-specific mule shoe and rupture-disk sub, 3 1/2-in. sacrificial tubing, along with required crossovers/adapters, ran in hole with a combination string 5-in drill-pipe and 3 1/2-in. tubing. To help with the field deployment, a dual-chamber top-drive cement head was also included into the tally.

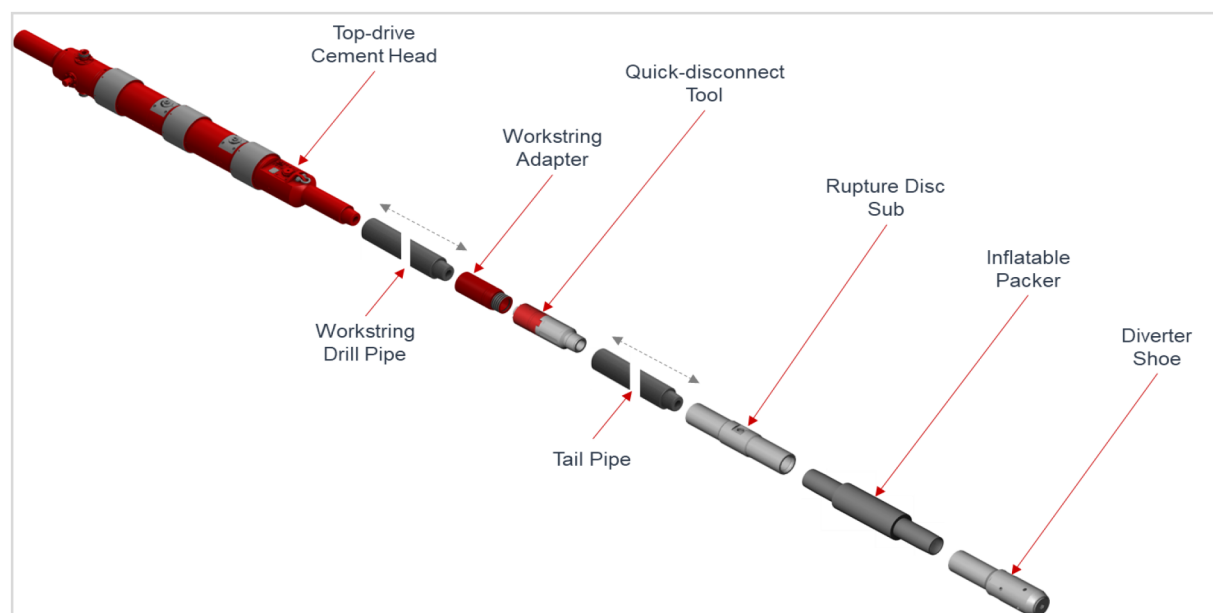


Figure 4—Detailed schematic representing the composition of string for an improved P&A

During execution, the inflatable packer was run to 2,846 m, with 300-m tail pipe separating the packer from the disconnection point, positioned at 2,546 m. Once at the final depth, packer inflation operating dart was released from top drive cement head and displaced down to the dedicated mule-shoe, where it provided required shut-in condition to hydraulically operate the inflatable packer, that was set and confirmed with a pressure hold of 1,400-psi. With indications of successful isolation of the wellbore downstream from the

packer, pressure was staged up to 2,200-psi, to rupture the frangible disk on circulation sub, positioned immediately above the packer.

With circulation achieved through the sub, cementing job activities started with pre-job well conditioning, followed by pumping planned cementing fluids (spacer and cement slurry). To initiate fluids displacement, disconnection operating dart was released and displaced from cement head down the string with SOBM. Once dart landed on its dedicated seat, it provided required shut-in condition to increase pressure inside the string, such to operate the disconnect mechanism, at 2,600 psi (disconnect pressure).

As a result, a loss of 15 kips of string weight was observed, along with circulation back to return lines, indicating successful separation of the tail pipe and placement of a 200-m long cement plug above a mechanical barrier (inflatable packer). An additional 250-m long cement plug was placed to cover the open hole-to-previous casing transition, still utilizing the retrieved portion of workstring from the previous job, consolidating the single-string, single-run P&A approach.

It is estimated that with efficiencies gained, this P&A resulted in USD 1.5 million in rig time only, allowing operator to complete activities ahead of schedule.

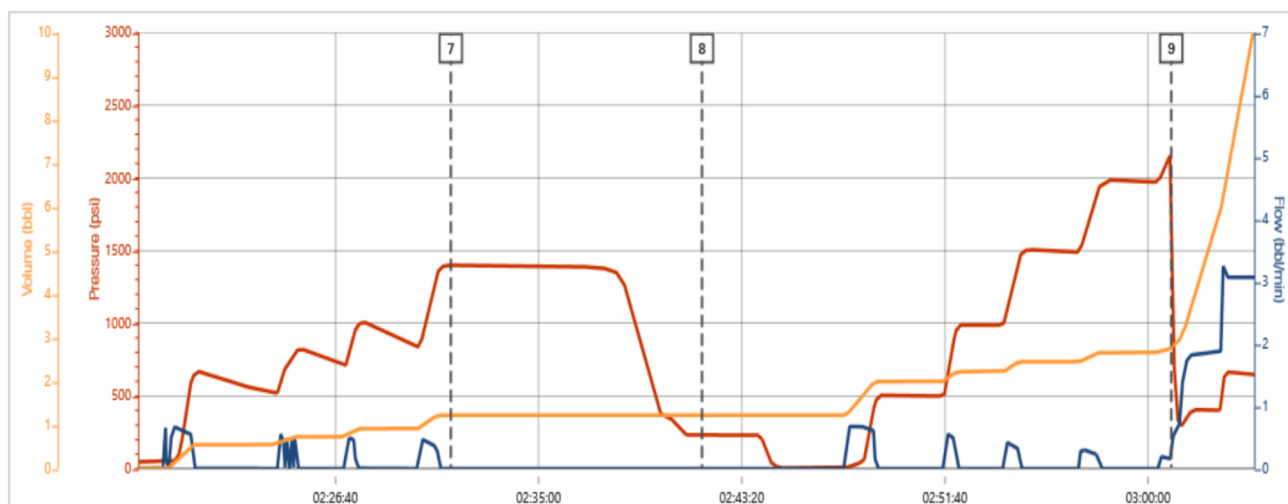


Figure 5—Sequence of events: Inflatable packer setting and Circulation through sub

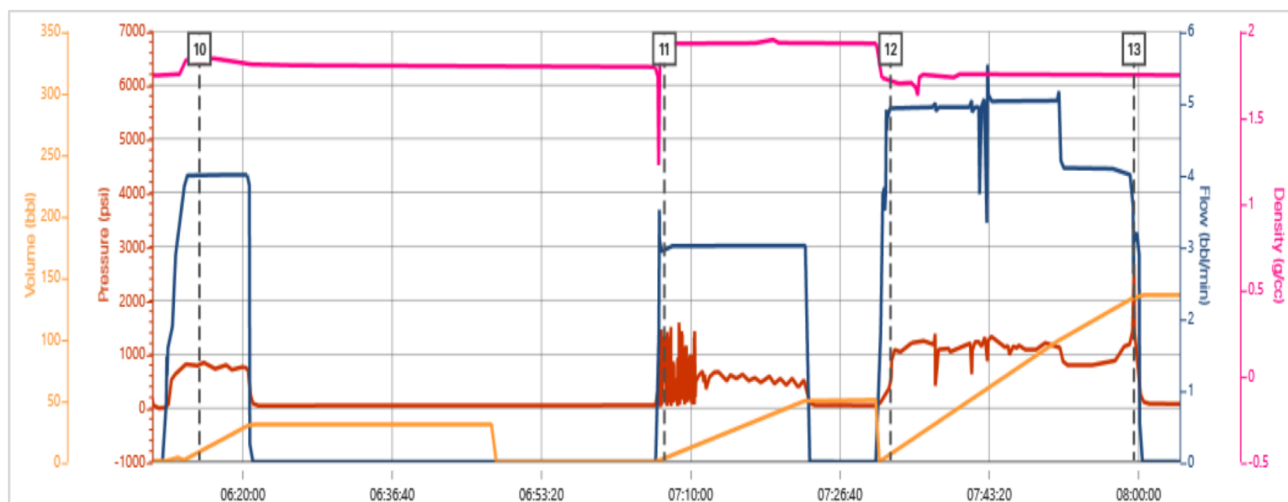


Figure 6—Sequence of events: Pumping of spacer, cement slurry and displacement, and String release

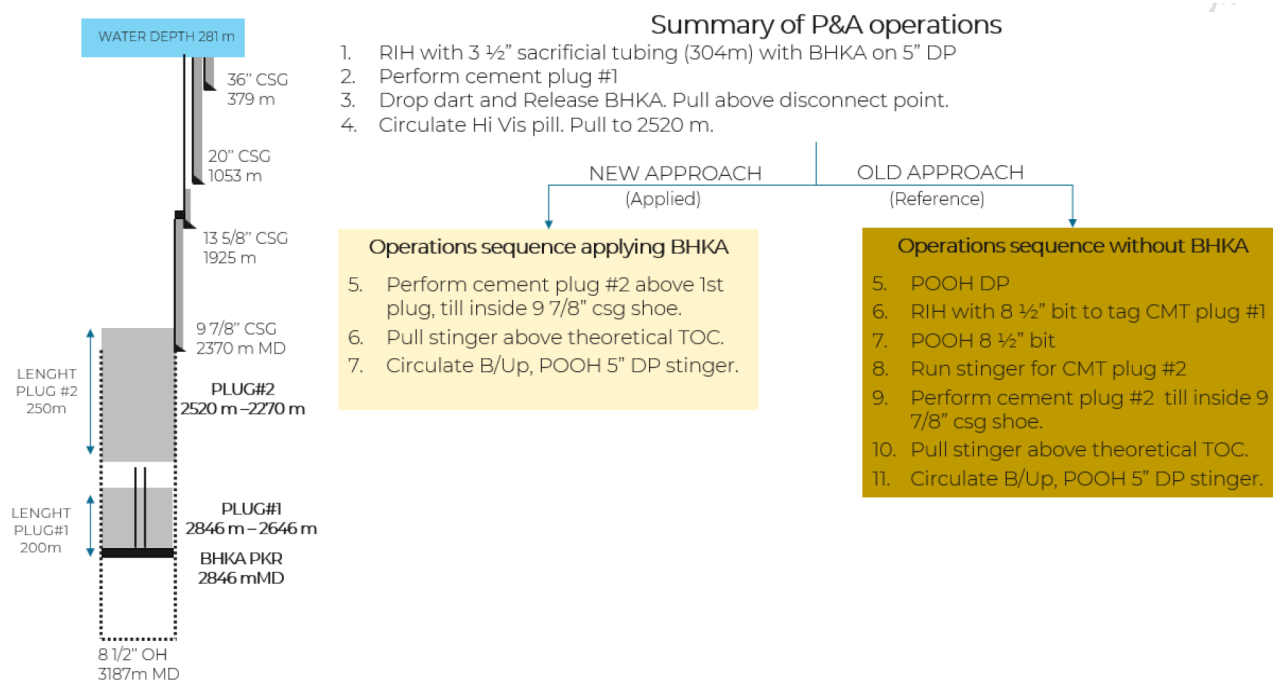


Figure 7—Summary of P&A operations

Case Study #2 – Africa

This second case study, which took place in West Africa region, a deepwater offshore environment application, which provides another valuable example of how the string quick-disconnect tool can significantly improve P&A operations across diverse fields.

The primary focus was on placing two cement plugs: one at the bottom of the well (Plug #1), and the second, a kickoff plug (Plug #2), to complete the well abandonment. Similar to the case in the Americas, conventional P&A method would have required multiple trips into the well to mechanically spot and tag the first cement plug and then proceed to place the second one. This would involve additional tools, more rig time, and greater potential for operational delays. To streamline operations, the proposed solution was planned to simplify cement plug setting procedures, as the system replaces consecutive stacked plugs to abandon and kick off with a single, longer cement plug.

For this particular well, to plug and abandon the 12 1/4-in. pilot hole, the operation involved running a 4 1/2-in. sacrificial tubing section, the disconnection tool, and 5-in. drill pipes used as landing string.

The operation was executed in adherence to the cement program. The cementing services team deployed the operating dart from the top drive cement head and displaced it to land and operate the single-collet release mechanism at a differential pressure of 2,200 psi applied on the landed dart. The objective of providing 100 m of isolation above the target zone was achieved on the first attempt.

The solution enabled the operator to complete the entire P&A in fewer trips, reducing rig time and reducing overall operational cost. It also ensured precise and effective cement plug placement, while eliminating the need for tagging the cement plugs, making operations simpler. It is estimated that, overall, rig time saving added up to approximately 40-45 hours.

The results from this application further solidify the single-trip quick-disconnection system as a highly effective tool for optimizing P&A operations. The system's ability to reduce rig time, increase cement plug integrity, and enhance operational safety makes it a valuable asset in well abandonment. Based on the success of this case study, Operators plan to continue replicating the approach in future projects to further refine and improve our P&A procedures.



Figure 8—Sequence of events: Pumping of spacer, cement slurry and displacement, and String release

Conclusion

The introduction of a BHA that is modular/based in Plug and Abandonment (P&A) strategy marks a significant step forward in well abandonment technology, offering a safer, faster, and more cost-efficient alternative to conventional methods. Traditional P&A operations often rely on multiple trips, staged cement placement, and mechanical tagging, all of which increase operational complexity, rig time, and exposure to safety risks. By combining a quick-disconnect tool, inflatable packer, and dedicated circulation sub into a single-trip modular system, the proposed approach eliminates many of these inefficiencies.

Field deployment across two distinct operating environments—offshore wells in the Americas and West Africa—demonstrated the system's flexibility, adaptability and robustness. In the first application the technology enabled the sequential placement of multiple cement plugs in a single trip, avoiding the need for intermediate tripping or wait-on-cement periods. In the second application the disconnecting tool allowed precise placement of both bottom and kickoff plugs without mechanical tagging, reducing rig time by an estimated 40–45 hours. In both cases, the resulting operational efficiency translated into cost savings on the order of USD 1.5 million per well.

The modular design of the system provides operators with a high degree of flexibility to tailor plug configurations for specific wellbore geometries, abandonment objectives, and regulatory requirements. The inflatable packer delivers a secure mechanical base that supports long cement columns while isolating weaker formations, thereby reducing the risk of losses and Equivalent Circulating Density (ECD) complications. The quick-disconnect tool ensures that the cement slurry remains undisturbed during release, preserving plug integrity and improving the likelihood of long-term zonal isolation.

Beyond the operational efficiencies, the presented methodology delivers significant HSE benefits. The single-trip approach minimizes personnel exposure to repetitive tripping operations and reduces overall rig time, lowering the probability of non-productive time (NPT) incidents. The improved precision in plug placement also reduces the likelihood of remedial work, further enhancing safety and cost-effectiveness.

By replacing the P&A procedure from multiple balanced plugs to a single cement job operation, the disconnect tool help minimize the volume of cement spacer/flush prepared then pumped ahead and behind the cement slurry. These spacers are usually circulated out of the well, resulting in mud contamination that does not have to be treated or disposed of. This benefit is even more relevant in the case of stacked cement plugs placed in long P&A intervals, where excess cement is also generally circulated out.

The successful trials not only validate the system's technical feasibility but also highlight its potential for broad application across a range of P&A scenarios. Its proven ability to operate effectively in both open-hole and cased-hole environments, as well as in vertical and deviated wells, makes it a versatile solution for

operators worldwide. Furthermore, the system's compatibility with varying well abandonment regulations allow for seamless integration into global decommissioning programs.

Looking ahead, the technology could be further enhanced through integration with real-time downhole monitoring and automation, enabling operators to optimize plug placement dynamically and respond to unexpected wellbore conditions without additional trips. Additional field trials in more challenging environments—such as ultra deepwater, high-pressure/high-temperature (HPHT) wells, or highly depleted formations—would expand the system's performance envelope and increase operator confidence in its application.

In conclusion, the modular-based BHA for P&A has proven to be a game-changing innovation in well abandonment operations. Its combination of operational simplicity, technical reliability, and economic benefit positions it as a best-practice solution for future abandonment campaigns. By reducing rig time, improving plug integrity, and enhancing safety, the system addresses critical industry challenges and sets a new benchmark for efficient and sustainable well decommissioning worldwide.

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System Of Units

PRESSURE	: pounds per square inch (psi)
DENSITY	: grams per cubic centimeter (g/cc)
FLOW	: barrels per minute (bbl/min) or (B/M)
FORCE	: kilopound force (kip)
LENGTH	: meters (m)
TEMPERATURE	: Celsius (°C)
VOLUME	: barrels (bbl)
DRILL PIPE AND	: inches (in.)
CASING SIZE	

Acronyms & Abbreviations

BHA	: Bottom hole assembly
ECD	: Equivalent Circulating Density
HP/HT	: High Pressure and high pressure
HSE	: Health, Safety and Environment
NPT	: Non-Productive Time
MW	: Mud Weight
P&A	: Plug and Abandonment
POOH	: Pull out of the hole
WBE	: Well Barrier Envelope
WOC	: Wait on Cement
SOBM	: Synthetic Oil Based mud
TD	: Target Depth

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